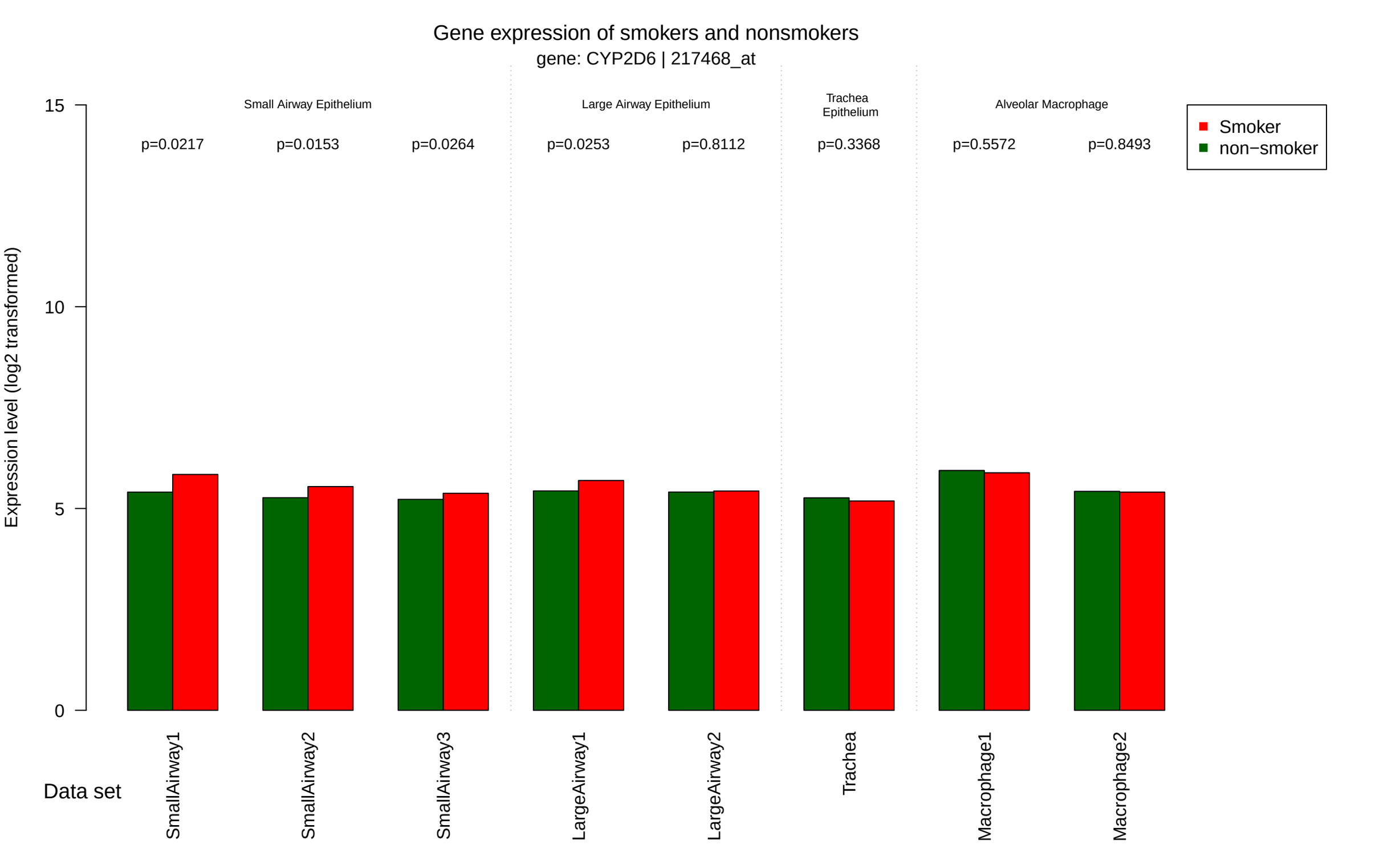
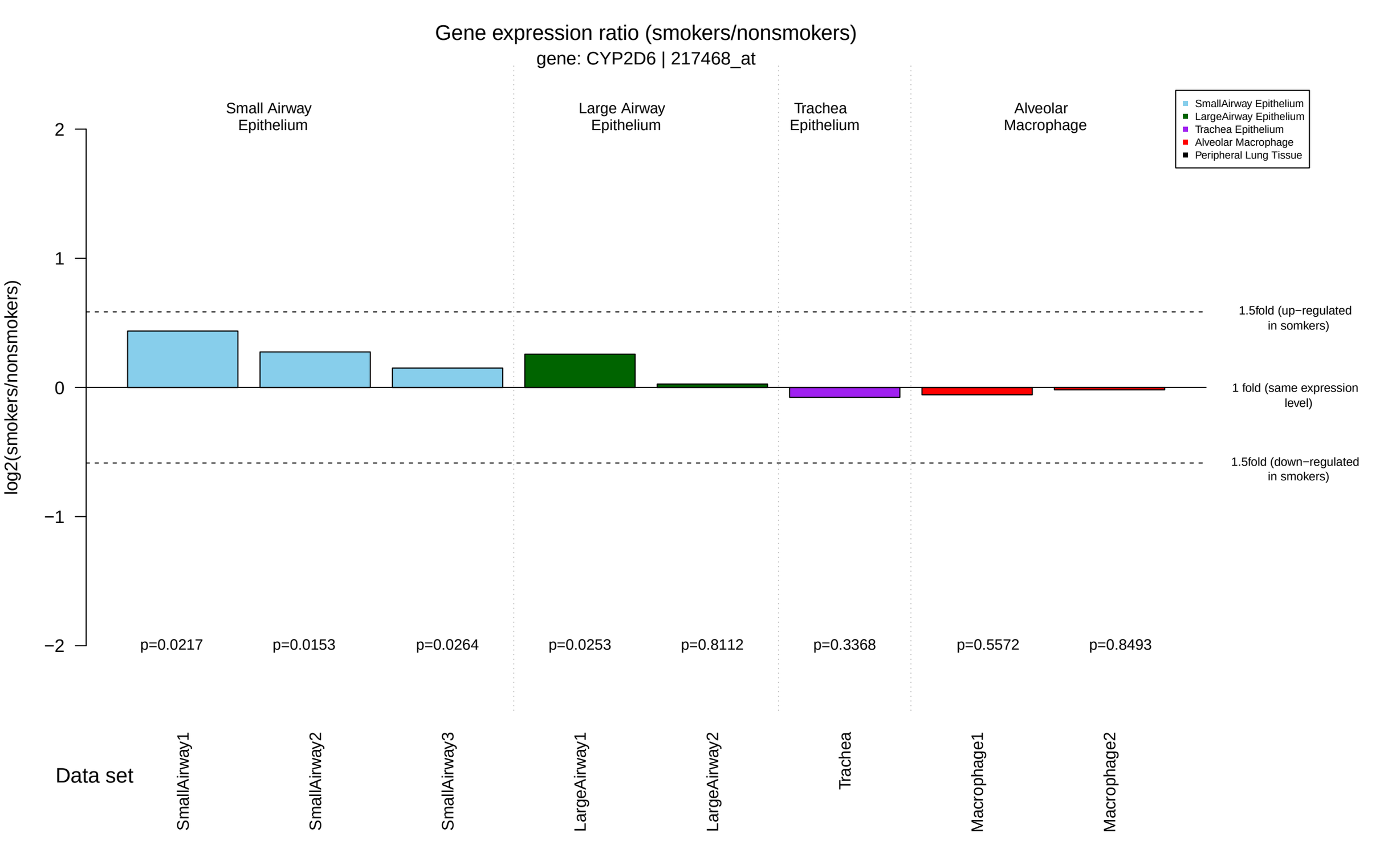
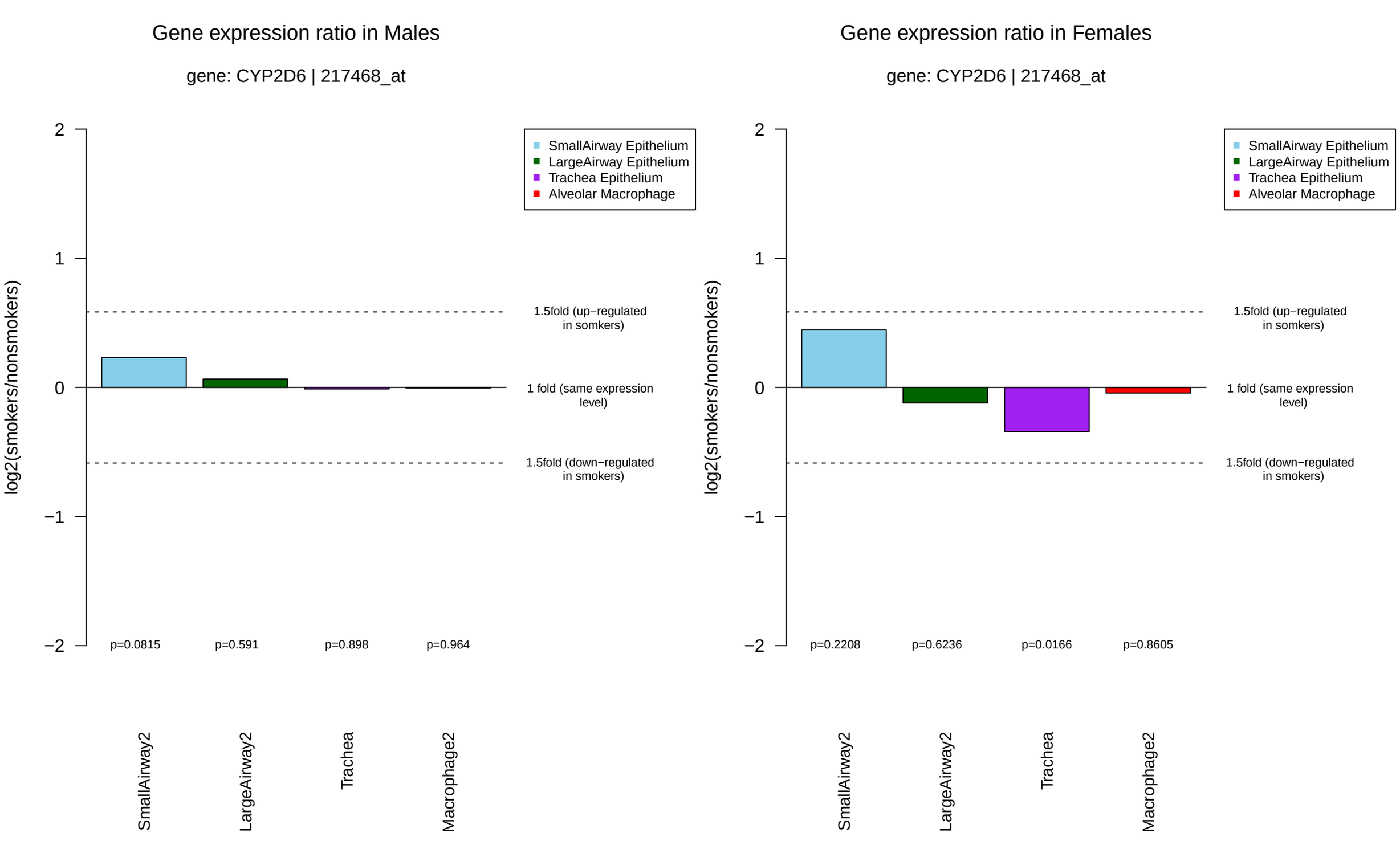
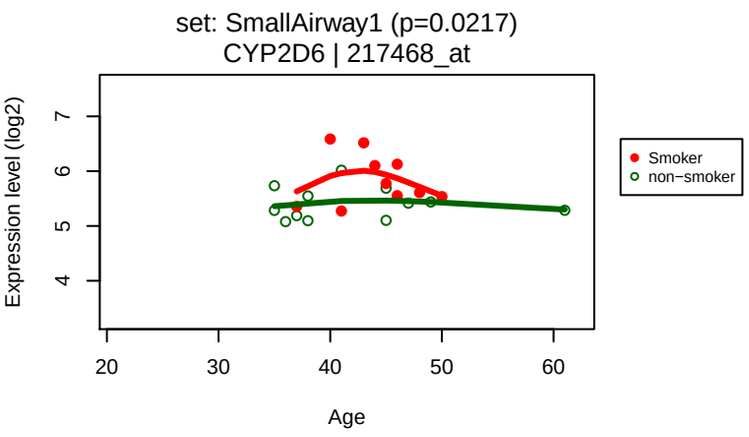
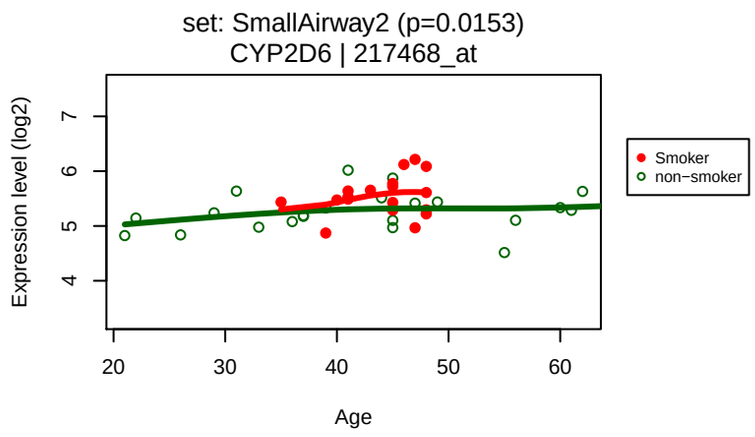
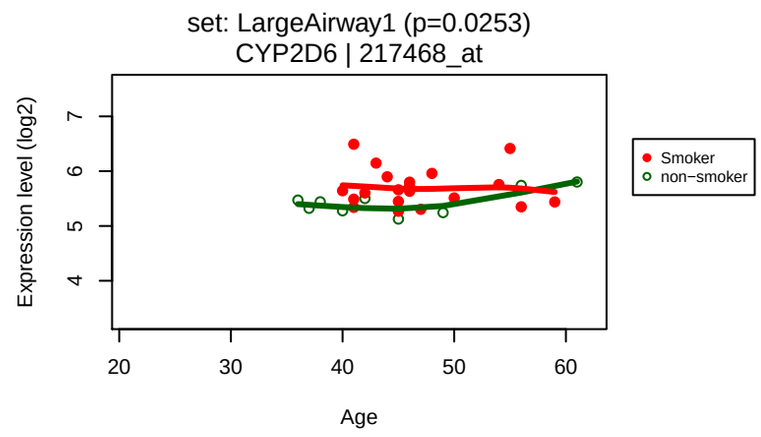

Gene expression of smokers and nonsmokers
gene: CYP2D6 | 217468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0217
p=0.0153
p=0.0264
p=0.0253
p=0.8112
p=0.3368
p=0.5572
p=0.8493
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

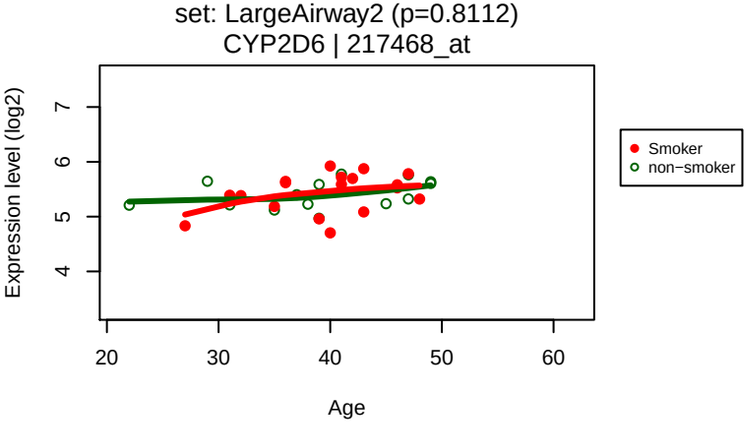

Gene expression ratio (smokers/nonsmokers)
gene: CYP2D6 | 217468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0217
p=0.0153
p=0.0264
p=0.0253
p=0.8112
p=0.3368
p=0.5572
p=0.8493
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

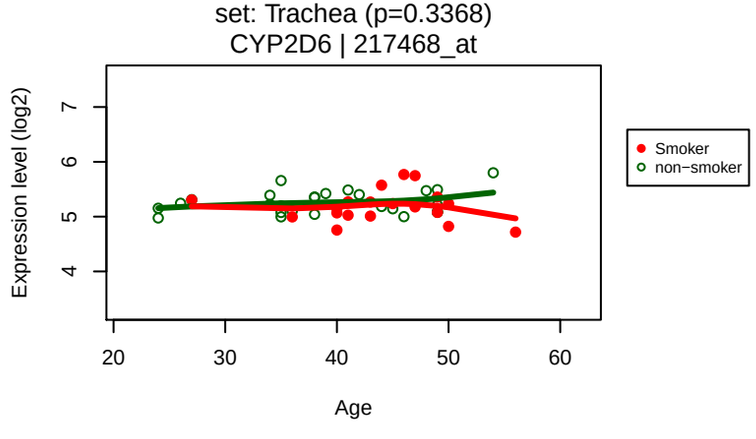

Gene expression ratio in Males
gene: CYP2D6 | 217468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0815
p=0.591
p=0.898
p=0.964
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2D6 | 217468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2208
p=0.6236
p=0.0166
p=0.8605
SmallAirway2
LargeAirway2
Trachea
Macrophage2

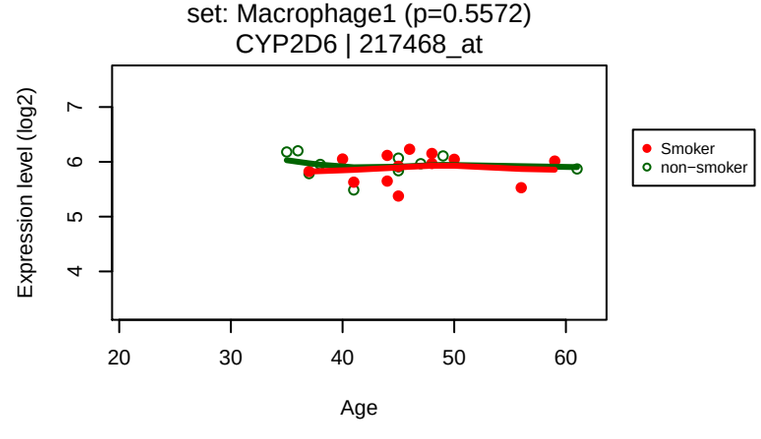

set: SmallAirway1 (p=0.0217)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker

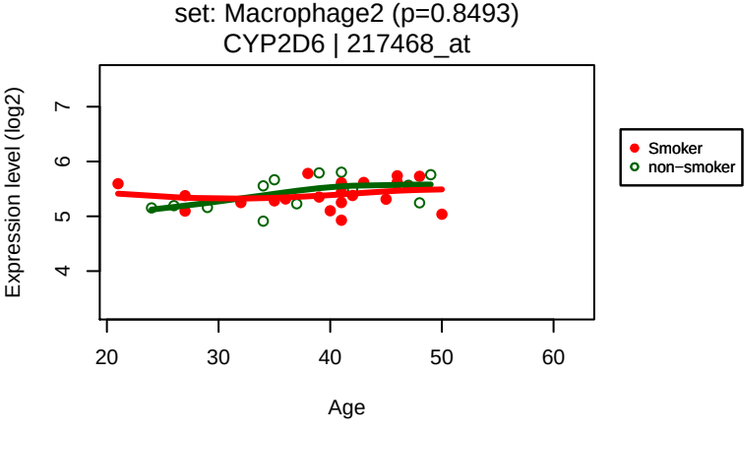

set: SmallAirway2 (p=0.0153)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker

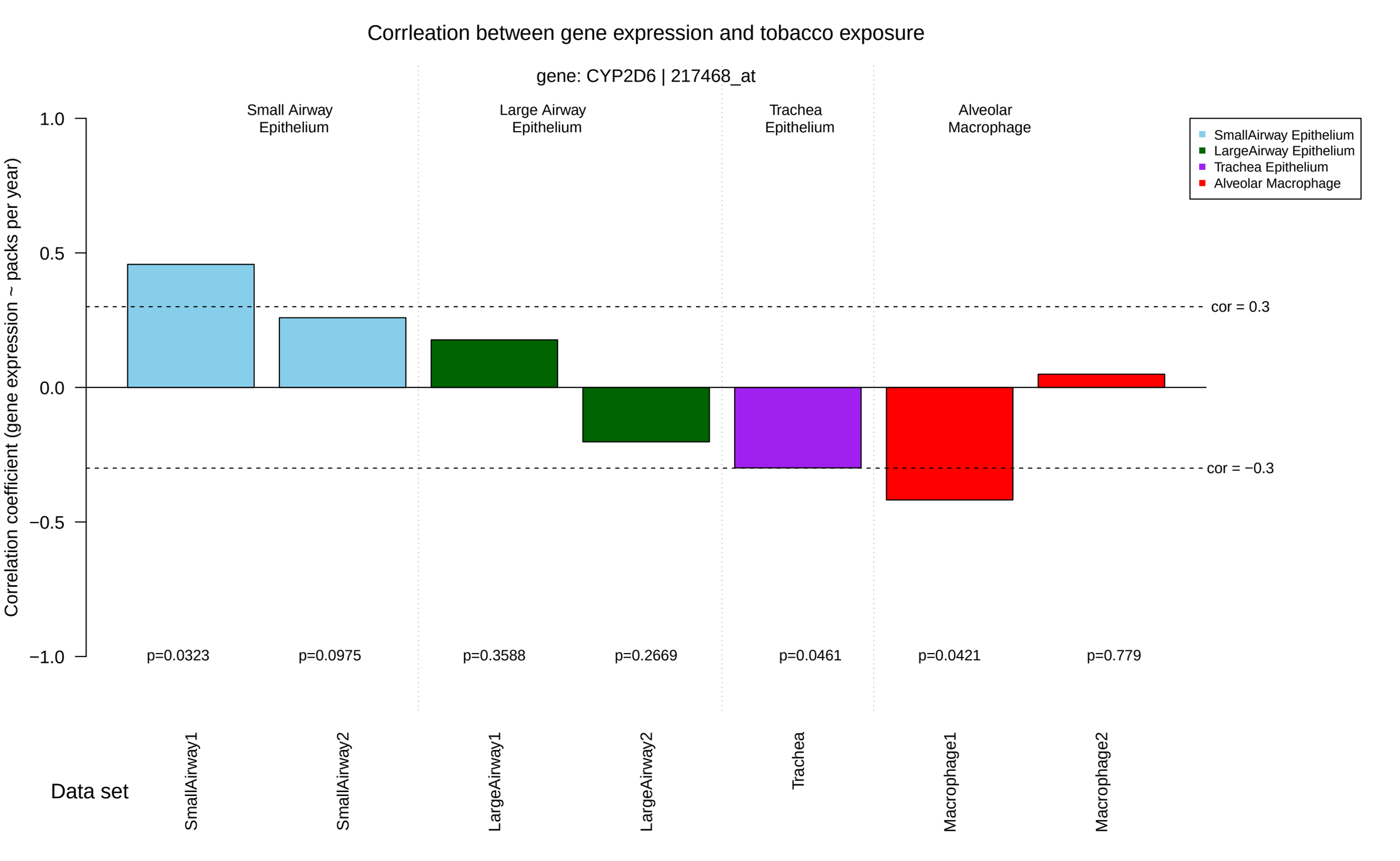

set: LargeAirway1 (p=0.0253)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8112)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3368)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5572)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8493)
CYP2D6 | 217468_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2D6 | 217468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0323
p=0.0975
p=0.3588
p=0.2669
p=0.0461
p=0.0421
p=0.779
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2